

Asymmetric VR Chores: A Social Presence Preliminary Study

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ABSTRACT

VR applications focused on daily activities for health care, focusing on either physical or cognitive interventions, continue to gain momentum as a result of consumer-level devices. However, interactions are mainly focused on single user experiences, with little research conducted on VR and asymmetric gameplay, which refers to the fact that players play the game using different media. This article presents a preliminary study focused on how the addition of asymmetry to a house chores VR game positively influences social presence when playing the game. The preliminary results indicate that the participants experienced high behavioral engagement when playing with another person.

Index Terms: Human-centered computing—Human computer interaction—Interaction paradigms—Virtual reality Human-centered computing—Human computer interaction—Interaction paradigms—Collaborative interactions

1 INTRODUCTION

The current availability of off-the-shelf VR headsets has sparked the development of experiences based on daily activities. For example, games that simulate cooking and office interactions have gained popularity and inspired research to develop a greater understanding of how these tools allow studying behavior, cognitive decline, and motor skills for health care and training experiences [5].

VR experiences predominantly cater to single-player experiences in isolation. Although multiplayer VR is gaining momentum, VR players remain isolated from others, in particular when co-located with others in the same space. While some cooperative multiplayer VR experiences allow for local play, this may require players to take turns or have both players using a VR headset. To address this shortcoming of multiplayer interaction, we have chosen the concept of asymmetric gameplay, in which two players can interact or play together while using different mediums. For example, one player is in VR wearing a headset, while the other is not. The positive impact of asymmetric gameplay has been reported in applications for improving communication [8], social connectedness [4], and cooperative interactions [7]. The lack of research on asymmetric VR is an open problem that has led to the development of a framework to test and expand future research on the most beneficial interaction paradigms for co-located asymmetric VR [6].

This paper presents a preliminary study to understand whether the use of asymmetric gameplay enhances social presence when performing chores in VR that can be perceived as tedious in real life. Our decision to use asymmetric gameplay is grounded on the reported benefits to communication, social connectedness, and cooperation in co-located scenarios to overcome the isolation typical VR creates between the player and others around them.

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2 MATERIALS AND METHODS

A preliminary within-subjects study focused on understanding how asymmetry impacted social presence when playing a chores VR game was conducted. The asymmetric game is a cooperative two-player game with two roles: a 'VR player' who uses the VR headset and an 'external player' who does not wear a VR headset and facilitates gameplay by using printed guides. To minimize carryover effects, each player's role was randomized on a first-come, first-serve basis. After playing the game in each role, participants completed the Social Presence Games Questionnaire (SPGQ), the System Usability Scale (SUS), and the NASA Task Load Index (TLX). The SPGQ [2] was chosen for this study, as its three subscales measure empathy, negative feelings, and behavioral engagement. In conjunction with usability and cognitive load measures collected using the SUS [1] and the raw NASA TLX [3], in addition to open feedback. The user study was reviewed and approved by the Research Ethics Board of Ontario Tech University [#17409] on 7 July 2023.

2.1 Participants

A total of 10 participants, 8 male, 1 female, and 1 non-binary, volunteered to participate in the study. Of these participants, 8 were in the age range of 18-24 years and the remaining 2 were in the age range of 35-54 years. The frequency of use of VR HMDs varied among participants, with only three participants using a VR HMD more than once a month. Professional and/or educational backgrounds also varied, with 8 participants reporting having a computer science, engineering, networking and IT, and game development background, and two participants reporting a health science background, with one being a first-year game development undergraduate student.

2.2 Asymmetric Game Development

The development of the asymmetric VR game used interconnected design aspects including: the activities to follow, virtual room design, game elements, and the core game loop. The chores were selected and gamified by adding scores associated with task completion and time highlighted by audiovisual cues that align with best practices and game design principles. Unity, OpenXR, and Autohand Unity were used in conjunction with the Meta Quest 2 headset.

The virtual room consists of a bed, a shelf, a window, and a drawer. The virtual room does not present a photorealistic environment for the purpose of minimizing discrepancies between the level of visual detail of a simulated environment that has been shown to encumber cognitive resources and negatively impact learning outcomes. In addition to seeking harmony between the user's level of control and the realism of the graphics, the game elements are visually clear and simple. Since our setting was focused on personal hygiene, representations of dirt and grime became important elements related to the game. We believe that indicating dirt with simple cartoonish graphics, rather than realistic graphics, made it much more apparent and visually distinct.

The house chores require pick & place tasks for cleaning and organizing segments of the room using VR controllers in seated or standing mode. Fig. 1e shows an external and VR player playing the game, including: i) Cleaning a window and a surface (Fig. 1a and Fig. 1b respectively), ii) Organizing a shelf (Fig. 1d), and iii) Disposing a can as shown in Fig. 1c.

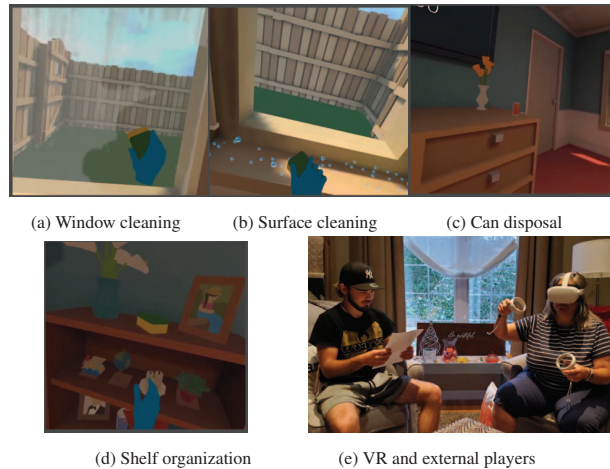


Figure 1: Implemented house VR chores.

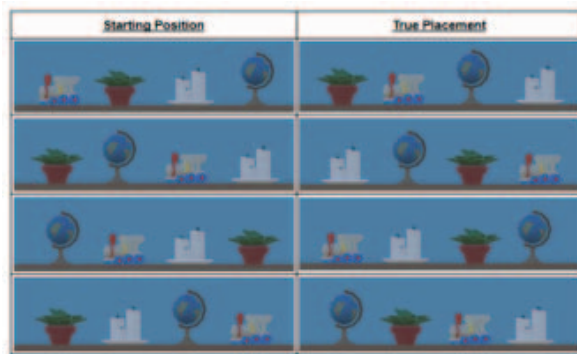


Figure 2: Sample guide available to the external player.

The VR player is tasked with communicating visual information about three visual elements within the room: a red can, trinkets on a bookshelf (sample shown on 2, and the contents of an abstract painting). Since external players cannot see what the VR player sees, they will need to interpret the information being communicated to determine the specific details to be used with a look-up sheet containing the necessary information to complete the game.

3 PRELIMINARY RESULTS

The SUS score for VR players was 89.5, compared to 88.2 for external players. Playing the game in either role was perceived as usable since usability scores above 68 are indicative of this [1]. Regarding cognitive load, the raw NASA TLX indicated that external players found the experience less demanding with a score of 26.1 compared to VR players, who found it slightly more demanding with a score of 28.6. Finally, the SPGQ subscale results, which are on a scale of 0-4, indicated that none of the participants experienced negative feelings for either role. However, the empathy subscale had VR players feeling less empathetic, with a score of 3.14 than the external players, with a score of 3.43. Regarding behavioral engagement, the subscale score was lower for VR players with 3.15 and higher for external players with 3.43.

4 CONCLUSION

This paper presented a preliminary evaluation of social presence in an asymmetric house chores VR game with participants playing

the game as VR and external players. Preliminary results, although lacking statistical significance due to the small sample size, provide valuable insight into how players perceive the game from different roles and how asymmetric gameplay impacts usability, cognitive load, and social presence. First, the higher usability and lower cognitive load may be explained by the background of the participants, as most of them were familiar with VR. Despite some difficulties reported when interacting with objects, locomotion, and instructions, the overall experience was well received by participants who expressed an understanding of the potential of such an approach to further boost social presence. We believe that the impact of asymmetry on behavioral participation indicated that the actions of each player, regardless of the role, are influenced by the information provided by the other player. It is important to note that external players reported higher levels of behavioral engagement than those who played in VR. This unexpected result could be attributed to the methods used to obtain and communicate information to the other player. For example, VR players only needed to visually communicate information about the virtual environment, while external players had to interpret the visual information and determine the associated task to communicate back to the VR player. Based on our results, we hypothesize that the additional level of complexity for external VR players is responsible for the increased perception of behavioral engagement, although more data are needed to validate that hypothesis.

Future work will focus on larger user testing, the addition of hand tracking support for natural user interactions, the comparison between single player and online synchronous asymmetric interactions, and the analysis of performance metrics to explore the applicability of the game on serious games and healthcare applications.

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